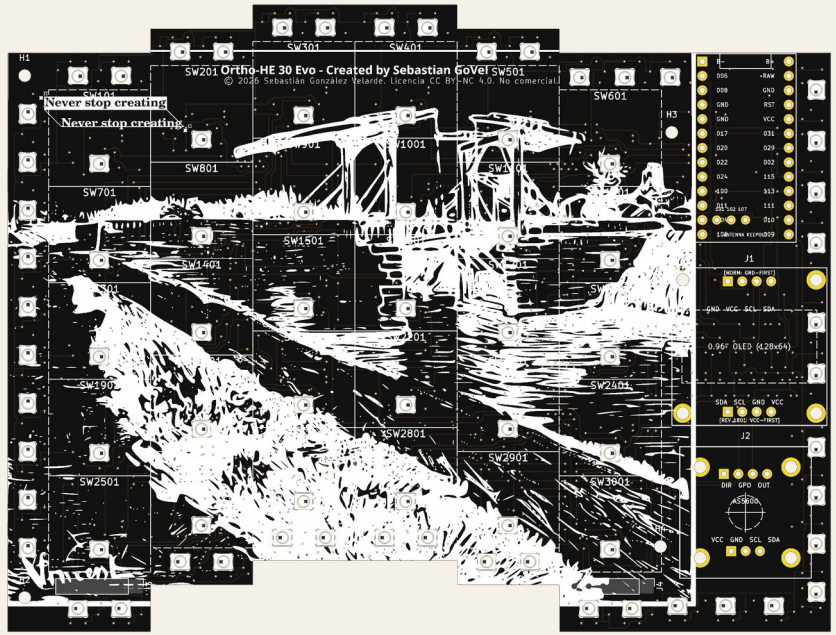
1517 FUND • MEDICI GRANT CANDIDATE

FloatForce

The Open-Source Continuous Analog Ergonomic Keyboard

Where every key floats on magnets instead of metal springs—turning physical press depth into expressive music, 360° analog game vectors, and zero-app precision mouse emulation.

Sebas GV • Jalisco, Mexico • Independent Hardware Maker
Open Hardware License: CERN-OHL-W v2 • Showcase: sebas-govel.github.io/HE_KB_Showcase



THE PROBLEM & OPPORTUNITY

Disrupting a Stagnant Multi-Billion Dollar Industry

1. Commercial Black Boxes

• Multi-hundred dollar price tags on closed consumer peripherals.
• Commercial Hall-effect switches cost exorbitant prices on AliExpress ($1.00+ each) with zero supply transparency.
• Black box firmware locked behind intrusive cloud desktop software.

2. No Design-Stage Modding

• Commercial switches force consumers into post-purchase 'modding' (lubing, filming, spring swaps) instead of customizing at the design stage.
• Fixed factory travel limits and single-magnet compromises.
• No adaptability for alternative tactile force curves or multilingual layouts.

3. Artificial Tier Paywalls

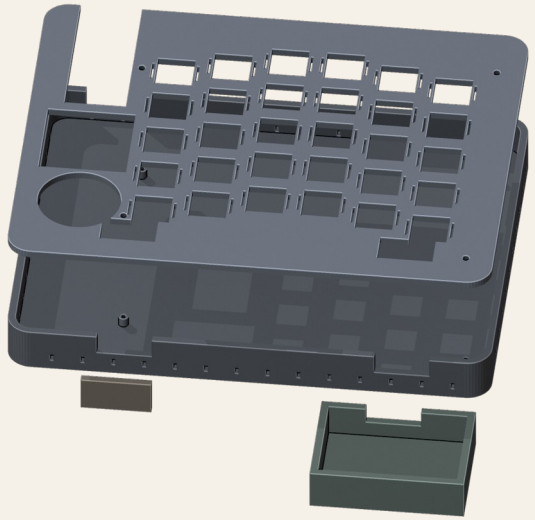
• Ergonomic health features (split layout, tenting, low-profile, wireless) are artificially segmented into $350+ luxury tiers.
• FloatForce's Mission: Full high-end analog sensing, 0°–25° drawbridge tenting, and split comfort—100% open-source under CERN-OHL-W v2 for anyone with an everyday 3D printer.

POSTURAL ERGONOMICS

Van Gogh's Drawbridge: Split & Tenting Architecture

Fixing 150 Years of Typewriter Flaws

• Drawbridge Elevation (0° → 25° Tenting): The outer pinky edges anchor firmly on the desk while the inner thumb edges rise, rotating forearms out of painful pronation into a neutral, natural handshake posture.
• Ergonomic 18 × 17 mm Pitch: Reduces finger travel distance by 15% compared to standard 19 mm grids, reducing ulnar deviation and carpal tunnel risk.
• Bottom-Mount Monolithic Canvas: All 30 Hall sensors, multiplexers, and power-gating circuitry reside on the bottom layer. The top surface is a monolithic tactile canvas featuring Van Gogh's 1888 Langlois Drawbridge etched in Matte Black and ENIG.

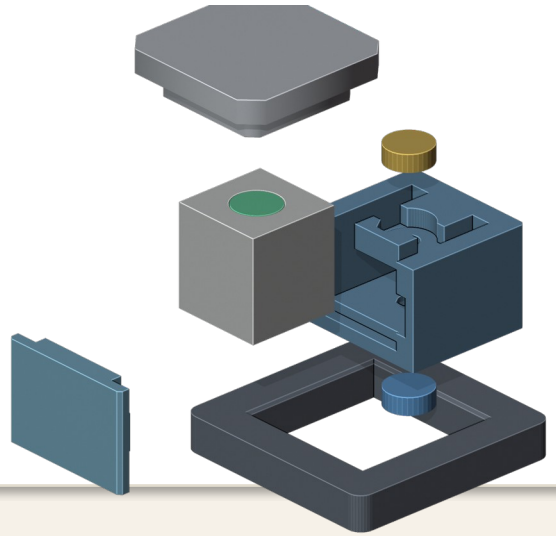


KINEMATIC HARDWARE

FloatForce Switch V5.0: Springless Magnetic Sensing

Springless 3-Magnet Levitation

• 2.36 mm Travel (+33% over V4.0): Sub-millimeter calibrated sockets (Ø4.06 mm / Ø4.16 mm) with knockout ejection pins.
• Monolithic U-Body + Keyed SideWall: Clamped by 3.0 mm plate; zero stem wobble and 0.000 mm^3 collision margin.
• Pre-rotated -45° in Z: Layer lines print 100% parallel to stroke, eliminating friction without requiring lube.



4 Continuous Analog Interaction Modes

• 1. Expressive MIDI Synth: Continuous velocity and filter swell directly on the home row keys.

• 2. WASD 360° Joystick: True proportional vector movement & stealth pacing in games.

• 3. Rapid Trigger & Dual-Stage: Dynamic reset upon lift; light tap for letter 'a', deep press for screenshot.

• 4. Precision Analog Mouse Emulation: PLANNED FIRMWARE: Cursor velocity proportional to key depth. Multi-speed navigation and CAD micro-positioning across any OS without third-party apps.

DECENTRALIZED INNOVATION

AI as an Engineering Copilot & Global Accessibility

A Solo Maker in Jalisco, Mexico

Historically, building custom analog hardware required a venture-backed team in Silicon Valley.

By leveraging modern AI as an engineering copilot:
- Magpylib for magnetic flux simulations
- KiCad 10 for multi-channel PCB routing
- Python for procedural CAD generation
- Zephyr RTOS for deterministic firmware

I designed, tested, and sent to fabrication this entire physical stack as an independent builder in Mexico.

Global Maker Accessibility

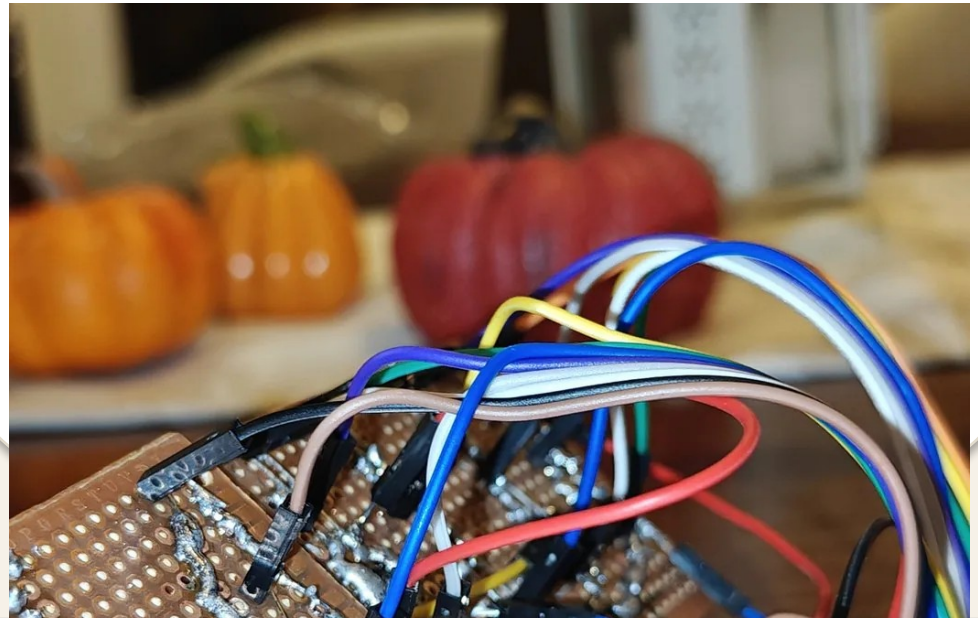
- 100% Support-Free FDM Printing: Switches and keycaps engineered to print with zero non-removable supports on everyday desktop 3D printers (Bambu A1, Ender 3, etc.).
- Accessible for All Alphabets: Standardized parametric CAD enables makers worldwide—from Latin America to Japan and Africa—to manufacture and customize localized keyboards at minimal cost.

ENGINEERING RIGOR

Physical Laboratory Proof: What's Real Today

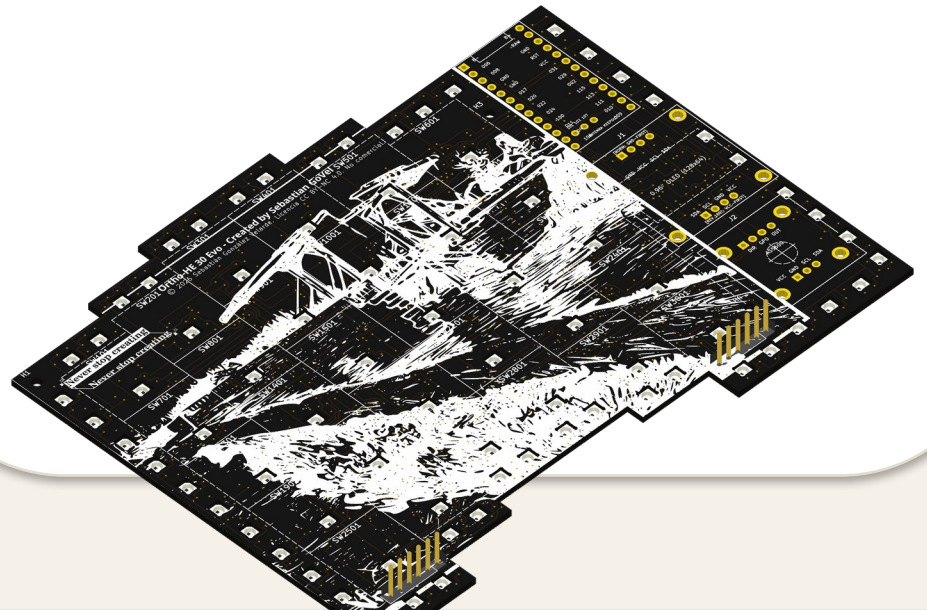
The 16-Key Functional Macropad

Bench-tested physical proof: 16 working FloatForce keys running Zephyr RTOS firmware. Validated magnetic repulsion, plate clamping, and zero mechanical binding under typing load.



PCB Rev A: In International Air Transit

SMT assembly complete at JLCPCB; currently in international air transit from Hong Kong to Mexico. Features 30x HAL403SO sensors, dual-bank TMUX1308 split (-34% capacitance), and 100 nF bypass caps.



THE MEDICI PITCH

$1,000 USD Grant Allocation & Sustainable Open Model

100% Forward-Looking Capital Runway

Past R&D (~$500 USD) was 100% self-funded from personal savings (not for reimbursement). The $1,000 USD grant funds forward development:

- $250.00 — Rev B Prototyping
  PCB fabrication run informed by Rev A testing, plus sensors, OLEDs, encoders, LiPo batteries & N52 magnets for working test units.
- $300.00 — Brand, Legal & Tax Advisory
  IMPI trademark registration in Mexico, prior-art search, and IP/tax advisory for certified 3rd-party builds.
- $450.00 — Materials R&D, Tooling & Creator
  Specialized filaments, hardened nozzles, BMCU multicolor/resin testing, and creator gear for build guides.

Sustainable Open Hardware Model

100% of the CAD files, schematics, and Zephyr firmware remain permanently open under CERN-OHL-W v2.

Financial sustainability is built on three independent pillars:
1. Trademark Certification (IMPI): Third parties can freely build hardware; certified builds pay a license for the official quality mark.
2. Official Patreon Community: Devlogs, early CAD drops, and participatory advisory.
3. Frictionless Direct Donations: GitHub Sponsors & community support.

FUTURE HORIZON

Beyond the Keyboard: The Future of Human-Computer Interaction

Chapter One of a Broader Interface Vision

- Frictionless Analog Ergonomic Mice: Magnetic Hall sensing applied to mouse buttons and scroll systems for pressure-sensitive panning and zero mechanical switch fatigue.
- Spatial Eye-Tracking Glasses with Monitor Bezel Detection: Lightweight glasses tracking pupil gaze while outward miniature sensors detect monitor bezels, translating gaze coordinates directly to the cursor across multi-screen setups without head turns.
- Bio-Subtle Continuous Inputs: Micro-movement accessibility interfaces blending human motor skill seamlessly with digital reality.

Thank you 1517 team for backing independent, unorthodox builders.

Sebas GV • Jalisco, Mexico • sebasgonzalezvelarde@outlook.com • sebas-govel.github.io/HE_KB_Showcase